

Qihang (Charlie) Wu

qihangwu@asu.edu • github.com/hahacharlie

RESEARCH INTERESTS

Electronic Design Automation (EDA), Computer Architecture, Heterogeneous Integration (HI) IC Design

EDUCATION

- Arizona State University, Ira A. Fulton Schools of Engineering**, Tempe, Arizona, United States
▪ M.S. in Electrical Engineering Started Jan 2026
- New York University, Tandon School of Engineering**, Brooklyn, New York, United States
▪ M.S. in Electrical Engineering Aug 2022 – May 2024
- University of Portland, Shiley School of Engineering**, Portland, Oregon, United States
▪ B.S. in Computer Science Aug 2015 – May 2020
▪ B.S. in Mechanical Engineering Aug 2015 – May 2020
- Sophia University**, Tokyo, Japan
▪ Exchange Student Jun 2017 – Aug 2017

PUBLICATION

- Qihang Wu and Austin Rovinski. 2025. **Revisiting Hardware Priority Queue Architectures**. Workshop Abstract. 2025 Open-Source Computer Architecture Research (OSCAR 2025) Workshop, Tokyo, Japan.
- Qihang Wu, Aman Arora and Vidya A. Chhabria. 2026. **CHICO-Agent: An LLM Agent for the Cross-layer Optimization of 2.5 D and 3D Chiplet-based Systems**. In Proceedings of the IEEE/ACM International Conference on LLM-Aided Design 2026.
- Zesong Jiang, Qihang Wu, Bing-Yue Wu and Jeff Zhang. 2026. **CAPO: Certification-Guided Agentic Workflow for Physical Design Parameter Optimization**. In Proceedings of the Great Lakes Symposium on VLSI 2026.
- Qihang Wu, Taizun Jafri, Aman Arora and Vidya A Chhabria. 2026. **VPR-Evolve: Multi-Agent-Driven Algorithm Evolution for FPGA Place and Route**. arXiv:2607.24998 [cs.AR]. <https://arxiv.org/abs/2607.24998>

PRESENTATION

- CHICO-Agent: An LLM Agent for the Cross-layer Optimization of 2.5 D and 3D Chiplet-based Systems. **Presentation** at the IEEE/ACM International Conference on LLM-Aided Design 2026, Stanford University, Stanford, CA, United States. Jul 2026
- Revisiting Hardware Priority Queue Architectures. **Presentation** at 2025 Open-Source Computer Architecture Research Workshop, Waseda University, Tokyo, Japan. Jun 2025
- NASA Lunabotics 2020 (University of Portland - Robotics). **Poster** at 2019 Oregon NASA Space Grant Consortium, Oregon State University, Corvallis, OR, United States. Nov 2019

RESEARCH EXPERIENCE

Revisiting Hardware Priority Queue Architectures

- Research Assistant (supervised by [Dr. Austin Rovinski](#)) May 2024 – May 2025
 - Designed and implemented multiple hardware priority queue architectures (Register Tree, BRAM Tree, Systolic Array, Hybrid Tree) on **Xilinx FPGA** using **SystemVerilog**, targeting scalable hardware accelerator applications
 - Developed a parameterized RTL design with a **unified testbench**, enabling architecture-agnostic benchmarking and verification
 - Conducted **timing analysis**, **resource utilization evaluation**, and **scalability testing** in Xilinx Vivado, identifying bottlenecks for different workloads
 - Investigated **area-latency trade-offs** across different architectures, optimizing queue throughput for hardware-based scheduling and networking applications

High-Efficiency SRAM Design for IoT Devices

- Student Research Assistant Intern (supervised by [Dr. Azeez Bhavnagarwala](#)) Sep 2023 – Mar 2024
 - Designed a 4kB SRAM array using **ASAP7** 7nm PDK in Cadence Virtuoso, focusing on low-power and high-density integration for IoT applications
 - Optimized **wordline and bitline decoder sizing**, improving **read/write access latency** and **power efficiency** under process variations
 - Conducted **Monte Carlo** simulations, ensuring robustness to **process-voltage-temperature (PVT)** variations
 - Performed **parasitic extraction** and **post-layout simulations**, verifying functional stability and leakage reduction techniques

SRAM Design Against Side-Channel Attacks

- Student Research Assistant Intern (supervised by [Dr. Azeez Bhavnagarwala](#)) Oct 2022 – Mar 2023

- Designed and synthesized an **AES-128** encryption/decryption hardware accelerator using **ASAP7 7nm PDK** in **Cadence Genus**, targeting **side-channel attack resistance**
- Developed a **gate-level netlist** and transistor-level **HSPICE power model**, enabling **power consumption analysis** for cryptographic security evaluation
- Simulated **Differential Power Analysis (DPA)** attacks on the AES block using **HSPICE transient simulations**, extracting power traces to assess cryptographic vulnerabilities
- Identified correlation between power fluctuations and AES internal operations, developing insights for countermeasure development in **secure SRAM design**

NASA Lunabotics Engineering Competition

- Team Member (supervised by **Dr. Ben Tribelhorn**) Sep 2019 – May 2020
 - Collaborated with a multidisciplinary team of 5 to develop key components of a Lunar Rover.
 - Implemented a **path-finding algorithm** that enables **autonomous navigation**.
 - Developed an **overlaying algorithm** and integrated it with a 2D Lidar unit on a rotating mount to assist in generating 3D scans of the surrounding terrain.

ACADEMIC PROJECTS

RISC-V RV32I Multi-Cycle Processor Implementation

Fall 2023

- Designed and implemented a 32-bit NYU-6463-RV32I processor using **Verilog**, supporting a subset of the **RISC-V RV32I** Instruction set, with a focus on **microarchitecture optimization** for **FPGA deployment**
- Conducted performance and area analysis, identifying **critical paths** and ensuring **synthesizability** and **optimal resource utilization** on FPGA
- Developed a comprehensive test suite, including **unit-tests** (ALU operations, branching, memory access) and **integration tests** (RC5 encryption/decryption, control flow validation), verifying correct execution at the instruction and system levels
- Implemented **hardware interfacing modules**, enabling interaction with **external peripherals** such as switches and LEDs, demonstrating real-world I/O operations
- **Synthesized** and deployed the design on a **Basys 3 Artix-7** FPGA Trainer Board, ensuring successful **timing closure** and **functional verification**

SRAM with BIST Engine

Spring 2023

- Designed and implemented a 256×4-bit SRAM using **SystemVerilog** in Cadence Virtuoso
- Verified SRAM functional correctness by integrating **SPICE-level** simulations with Behavioral RTL models
- Developed and embedded a **Built-In Self-Test (BIST) engine**, implementing industry-standard test patterns
- Synthesized the SRAM-BIST system using Cadence GENUUS, **optimizing for area, power and timing constraints**
- Performed **static timing analysis (STA)** in Synopsys PrimeTime ensuring timing closure

INDUSTRIAL EXPERIENCE

Artly AI, Seattle, Washington, USA

Jun 2020 – May 2022

- Hardware System Engineer
- *Artly AI (formerly known as Artly Coffee) is a Seattle-based robotic start-up developing an AI-powered robotic barista with the goal of automating and enhancing the coffee preparation process. Raised over **350 Million dollars** on [StartEngine](#).*
 - Created innovative and customized **hardware designs** for components that improve the system's stability and expand the robot's functionality.
 - Integrated high-precision scales (0.1g accuracy, 0.01g resolution) into the robotic system, **enhancing measurement accuracy by 80%**, improving beverage consistency.
 - Designed and built a motorized rotating and tilting platform; **achieved a 120% increase in latte art success rate**.
 - Developed **embedded firmware (C, Arduino, ESP8266, STM32)** for various components used for the robotic system interfacing with the robotic control system, improving hardware-software integration.

Hyster-Yale, Portland, Oregon, USA

- Mechanical Engineer Intern Jun 2018 – Mar 2020

ACHIEVEMENTS

PATENT

- "SYSTEM AND METHOD FOR ROBOTIC FOOD AND BEVERAGE PREPARATION USING COMPUTER VISION", PCT/US21/33430 May 2021

CERTIFICATION

- Oregon State Board of Examiners for Engineering, Engineer in Training (E.I.T.) Jun 2020

AWARD

- Undergraduate Team Experience Award – OGSC 2019 – 2020

ACADEMY

- Spring 2018 Dean's List May 2018